

DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to VTU, Belagavi - 590018)
Accredited by National Assessment & Accreditation Council (NAAC) with 'A' grade
Shavige Malleshwara Hills, Kumaraswamy Layout
Bengaluru-560111



BATCH NO- A5



Interdisciplinary Project (1BPRJ208)
Report on

“NEURO VP DHARA CORE”

Submitted in partial fulfillment for the award of degree of

Bachelor of Engineering
in
Electrical and Electronics Engineering

Submitted by

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JNANASANGAMA, BELAGAVI-590018

2025-26

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2025-2026

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

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DEPARTMENT OF CSE (IOT & BLOCKCHAIN)



CERTIFICATE

Certified that the Interdisciplinary project report entitled “**Neuro VP Dhara Core**” carried out by **Ayush Kumar (1DS25IC017)**, **Sourav Kumar (1DS25IC053)**, **Aniket V Patil (1DS25EE012)**, **K Satya Prakash (1DS25EE055)** are bonafide students of **DAYANANDA SAGAR COLLEGE OF ENGINEERING**, an autonomous institution affiliated to VTU, Belagavi in partial fulfillment for the award of Degree of **Bachelor of Engineering in Electrical and Electronics Engineering** during the year **2025-2026**. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of work prescribed for the said Degree.

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DECLARATION

We, Ayush Kumar (1DS25IC017), Sourav Kumar (1DS25IC053), Aniket V Patil (1DS25EE012), K Satya Prakash (1DS25EE055), hereby declare that the Interdisciplinary project work entitled “Neuro VP Dhara Core” has been independently done by us under the guidance of ‘Deekshitha Arasa’, Assistant Professor, Electrical & Electronics Engineering department and submitted in partial fulfillment of the requirement for the award of the degree of Bachelor of Engineering in Electrical & Electronics Engineering, at Dayananda Sagar College of Engineering, an autonomous institution affiliated to VTU, Belagavi during the academic year 2025-2026.

We further declare that we have not submitted this report either in part or in full to any other university for the award of any degree.

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Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru - 560 111. India

Department of Electrical and Electronics Engineering

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CO-PO Mapping

COURSE OUTCOMES

CO1: Identify and define problems requiring interdisciplinary knowledge.

CO2: Apply basic concepts of science, engineering and technology to design simple solutions.

CO3: Work effectively in teams with defined roles and responsibilities

CO4: Use project management, documentation and presentation skills.

CO5: Develop socially relevant, sustainable, and innovative prototypes/solutions.

CO-PO mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO8	PO9	PO10	PO11
CO1	2	1	1	1						1
CO2	2	1	1	1						1
CO3							2	2		1
CO4					1		2	2	1	1
CO5					1	2				1

ABSTRACT

Agriculture accounts for a significant portion of global freshwater consumption, making efficient water management essential for sustainable farming. Traditional irrigation practices often rely on manual observation and estimation of soil conditions, which can result in overwatering, underwatering, reduced crop productivity, and unnecessary water wastage. To address these challenges, this project presents Neuro VP Dhara Core (Smart Roots), an IoT-based intelligent soil monitoring and predictive water management system designed for small and medium-scale agricultural applications.

The proposed system utilizes an ESP32 microcontroller as the central processing unit, integrated with a Capacitive Soil Moisture Sensor and DHT22 Temperature and Humidity Sensor to continuously monitor environmental and soil conditions. Real-time sensor data is collected, processed, and transmitted to cloud platforms for remote monitoring and analysis. Based on predefined moisture thresholds, the system can automatically control a water pump through a relay module, ensuring efficient irrigation and reducing human intervention.

In addition to real-time monitoring, the system incorporates AI-assisted recommendations to support informed irrigation decisions and improve overall farm management. The integration of IoT technologies, cloud connectivity, and automation enables farmers to access live data, optimize water usage, and enhance agricultural productivity. The proposed solution is designed to be affordable, scalable, and suitable for deployment in resource-constrained farming environments.

The project demonstrates how smart sensing, automation, and data-driven decision-making can contribute to sustainable agriculture by minimizing water wastage, improving crop health, and supporting environmentally responsible farming practices.

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CHAPTER 1: INTRODUCTION

1.1 Background and Concept

Agriculture is one of the most water-intensive sectors, consuming nearly 70% of the world's freshwater resources. Efficient irrigation management is therefore essential for ensuring sustainable agricultural practices and maximizing crop productivity. However, many farmers still rely on traditional irrigation methods that depend on manual observation and estimation of soil conditions. These practices often result in overwatering or underwatering, leading to water wastage, reduced crop yield, and increased operational costs.

Recent advancements in the Internet of Things (IoT), embedded systems, and artificial intelligence have enabled the development of smart agriculture solutions capable of monitoring environmental conditions in real time. By integrating sensors, wireless communication, cloud platforms, and automated control systems, farmers can make informed decisions regarding irrigation and crop management.

The proposed project, **Neuro VP Dhara Core (SmartRoots)**, is an intelligent soil monitoring and predictive water management system that utilizes an ESP32 microcontroller, soil moisture sensing, temperature and humidity monitoring, cloud connectivity, and AI-assisted recommendations. The system continuously monitors soil and environmental conditions, provides real-time data visualization, and automatically controls irrigation based on soil moisture levels. This approach helps optimize water usage, improve crop health, and support sustainable farming practices.

1.2 Need for the Project

Agricultural productivity is highly dependent on proper irrigation and environmental management. Many small and medium-scale farmers lack access to affordable technologies that provide real-time information about soil and climatic conditions. As a result, irrigation decisions are often based on experience or assumptions rather than actual field data.

The major needs addressed by this project are:

- Lack of real-time soil monitoring systems for small-scale farmers.
- Excessive water wastage due to inefficient irrigation practices.
- Reduced crop productivity caused by improper soil moisture levels.

- Limited availability of affordable smart farming solutions.
- Increasing demand for sustainable and technology-driven agriculture.
- Need for automated irrigation systems that minimize human intervention.

The proposed system addresses these challenges by providing a cost-effective and intelligent monitoring solution capable of improving irrigation efficiency and resource management.

1.3 Problem Statement

Agricultural fields often suffer from inefficient irrigation practices due to the absence of continuous soil condition monitoring. Farmers frequently face difficulties in determining the optimal time and quantity of water required for irrigation, resulting in overwatering, underwatering, increased water consumption, and reduced crop productivity.

Furthermore, existing smart farming solutions are often expensive and difficult to deploy for small-scale farmers. Therefore, there is a need for an affordable, reliable, and intelligent monitoring system that can provide real-time soil and environmental data, automate irrigation decisions, and support efficient water management practices.

1.4 Objectives of the Project

The primary objectives of the project are:

1. To monitor soil moisture levels continuously using a capacitive soil moisture sensor.
2. To measure environmental temperature and humidity using a DHT22 sensor.
3. To develop an ESP32-based smart monitoring system with wireless communication capabilities.
4. To automate irrigation control through a relay-operated water pump based on soil moisture conditions.
5. To provide real-time monitoring and data visualization through cloud-based platforms.
6. To generate AI-assisted recommendations for better irrigation and agricultural management.
7. To reduce water wastage and improve overall agricultural efficiency.

8. To develop an affordable and scalable smart farming solution suitable for small and medium-scale farms.

1.5 Scope and Applications

The scope of this project extends to smart agriculture and precision farming applications where efficient water management is essential. The system can be deployed in agricultural fields, gardens, nurseries, and greenhouse environments to provide continuous monitoring and automated irrigation support.

Applications of the proposed system include:

- Precision agriculture and smart farming.
- Automated irrigation systems.
- Greenhouse monitoring and management.
- Home gardens and smart landscaping.
- Plant nurseries and horticulture farms.
- Agricultural research and experimental farms.
- Water conservation projects.
- Remote farm monitoring systems.

The modular architecture of the system allows future expansion through additional sensors, mobile applications, and advanced AI-based decision-making features.

1.6 Expected Outcome

The expected outcome of the project is the successful development of an IoT-enabled smart soil monitoring and irrigation management system capable of collecting, processing, and transmitting real-time agricultural data. The system is expected to:

- Provide accurate soil moisture, temperature, and humidity monitoring.
- Automatically activate irrigation when soil moisture falls below a predefined threshold.
- Reduce unnecessary water consumption and improve irrigation efficiency.

- Enable remote monitoring through cloud-based dashboards.
- Support farmers with intelligent recommendations for irrigation management.
- Improve crop growth conditions through timely watering decisions.
- Demonstrate the practical implementation of IoT and automation technologies in agriculture.

Ultimately, the project aims to contribute to sustainable agriculture by promoting efficient water utilization, reducing resource wastage, and enhancing farm productivity through smart technology integration.

CHAPTER 2: LITERATURE REVIEW

S. No.	Title of Work	Author(s)	Year	Key Findings	Limitations
1	IoT-Based Smart Irrigation System Using Deep Learning Techniques	Saranya et al.	2023	Utilized IoT sensors and deep learning algorithms for automated irrigation with high prediction accuracy.	High implementation cost and strong dependency on continuous internet connectivity.
2	GSM-Based Automated Irrigation Monitoring System	Veeramakali et al.	2020	Developed a low-cost irrigation automation system using GSM communication for remote monitoring.	Limited intelligence and operates primarily on predefined rules without predictive capabilities.
3	Random Forest-Based Crop Recommendation System Using Machine Learning	Apu et al.	2025	Applied machine learning techniques for crop recommendation with good prediction accuracy.	Lacks integration with real-time field sensors and automated irrigation control.
4	Smart Agriculture Platform (Fasal)	Fasal Technologies	Commercial	Provides AI-driven monitoring, weather prediction, disease alerts, and irrigation recommendations.	Expensive for small-scale farmers and requires subscription-based services.
5	IoT-Based Soil Moisture Monitoring System	Various Researchers	2022	Demonstrated real-time soil moisture monitoring using wireless sensor networks.	Focuses mainly on monitoring and lacks intelligent decision-making support.

Table 2.1 – Literature Survey

2.1 Summary of Literature Review

The reviewed studies demonstrate the growing adoption of IoT, machine learning, and automation technologies in modern agriculture. Most existing systems provide real-time monitoring, remote access, and irrigation control features. Some solutions also integrate artificial intelligence for predictive analysis and decision support.

However, many of these systems suffer from one or more of the following limitations:

- High implementation and maintenance costs.
- Dependence on continuous internet connectivity.
- Limited support for small and medium-scale farmers.
- Lack of integration between sensing, automation, and intelligent recommendations.
- Absence of affordable predictive irrigation support.

These limitations indicate the need for a cost-effective and scalable smart agriculture solution that combines real-time monitoring, automated irrigation, cloud connectivity, and AI-assisted decision-making.

2.2 Need for the Proposed Project (Research Gap)

Despite significant advancements in smart agriculture technologies, there remains a gap between sophisticated commercial solutions and the practical needs of small-scale farmers. Existing systems are either expensive, difficult to deploy, or lack intelligent predictive capabilities.

The proposed project, Neuro VP Dhara Core (SmartRoots), addresses these limitations by developing an affordable IoT-based soil monitoring and predictive water management system. The project integrates an ESP32 microcontroller, capacitive soil moisture sensing, temperature and humidity monitoring, cloud connectivity, and AI-assisted recommendations into a single platform.

The proposed system aims to:

- Provide real-time monitoring of soil and environmental conditions.
- Automate irrigation using moisture-based control logic.

- Reduce water wastage and improve irrigation efficiency.
- Offer AI-assisted recommendations for better farm management.
- Deliver a low-cost solution suitable for small and medium-scale farmers.
- Enable remote monitoring through cloud-based dashboards.

By combining sensing, automation, IoT connectivity, and intelligent decision support, the proposed system bridges the gap between traditional irrigation methods and advanced smart farming technologies, thereby contributing to sustainable agricultural development.

Agriculture is increasingly adopting automation and Internet of Things (IoT) technologies to improve productivity, resource management, and sustainability. Several researchers have developed smart irrigation systems that utilize sensors, microcontrollers, and wireless communication technologies to automate irrigation processes and monitor environmental conditions. These systems aim to reduce water wastage, minimize human intervention, and improve crop yield.

The following table summarizes some of the important studies related to automated irrigation and soil monitoring systems.

CHAPTER 3 : METHODOLOGY & DESIGN

3.1 Block Diagram

The block diagram illustrates the overall architecture of the Neuro VP Dhara Core (SmartRoots) system. The ESP32 microcontroller collects data from the capacitive soil moisture sensor and DHT22 sensor, processes the information, and controls the water pump through a relay module while transmitting real-time data to the cloud dashboard for monitoring and AI-assisted irrigation recommendations.

BLOCK DIAGRAM OF NEURO VP DHARA CORE (SMARTROOTS)

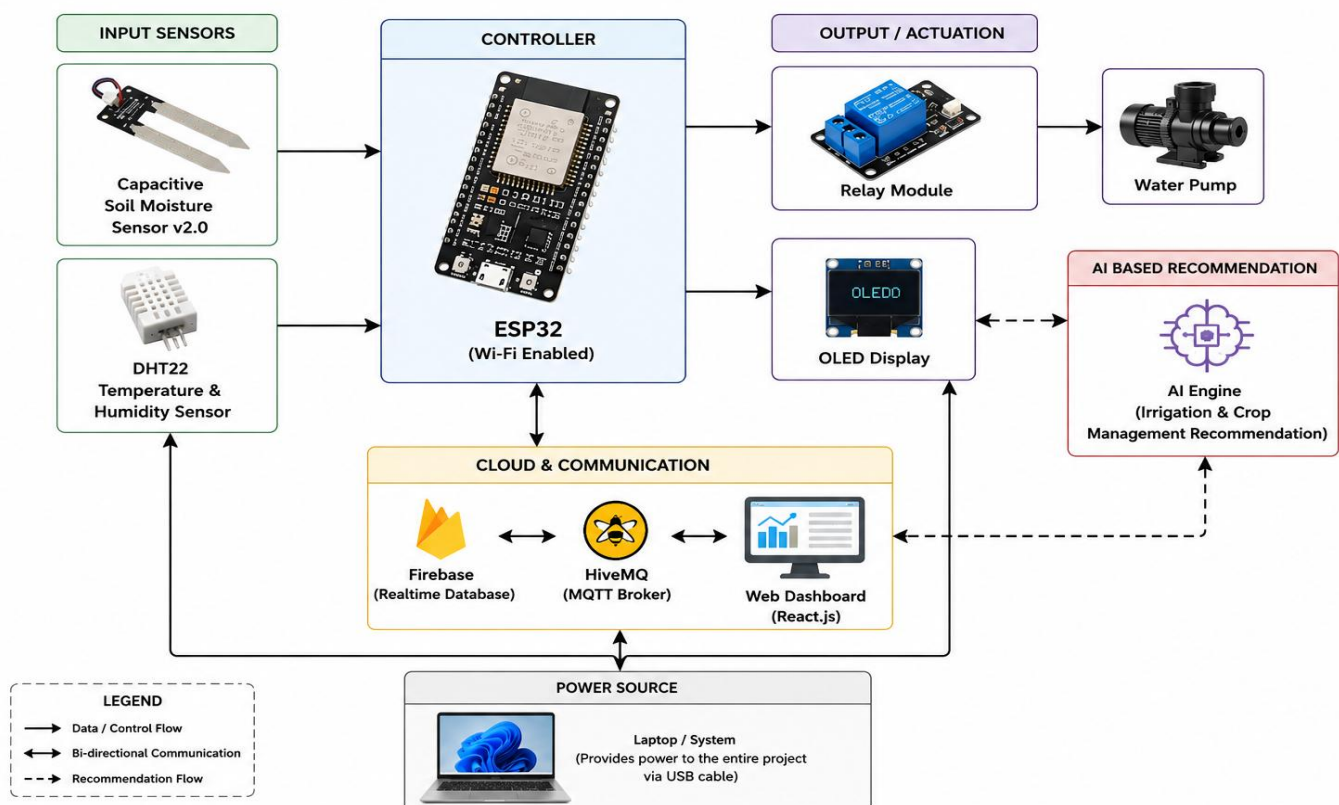


Figure 3.1 – Block Diagram

3.2 Working Principle

The Neuro VP Dhara Core (SmartRoots) system operates by continuously monitoring soil and environmental conditions using a Capacitive Soil Moisture Sensor and a DHT22 Temperature and Humidity Sensor. These sensors are interfaced with the ESP32 microcontroller, which acts as the central processing unit of the system. The entire prototype is powered through a USB

connection from a laptop, which also serves as the platform for programming, monitoring, and data visualization.

The soil moisture sensor measures the moisture content present in the soil, while the DHT22 sensor records the surrounding temperature and humidity. The ESP32 periodically acquires these sensor readings and compares the soil moisture value with a predefined threshold. If the soil moisture level falls below the threshold, the ESP32 activates a relay module, which switches ON the water pump to irrigate the soil. Once the moisture level reaches the desired range, the relay is deactivated, turning the pump OFF and preventing overwatering.

Simultaneously, the collected sensor data is transmitted through Wi-Fi to cloud platforms such as Firebase and HiveMQ for real-time monitoring and storage. The data is displayed on a web dashboard, enabling users to remotely observe field conditions. Based on the collected data, the AI recommendation module analyzes environmental trends and provides intelligent suggestions related to irrigation management and crop care. This integrated approach helps optimize water usage, reduce manual intervention, and improve agricultural productivity through smart decision-making.

3.3 : Hardware Components Used

Sl. No.	Hardware Component	Specification / Model	Purpose
1	ESP32 Development Board	ESP32-WROOM-32	Main controller for data processing and communication
2	Capacitive Soil Moisture Sensor	Analog Soil Moisture Sensor	Measures soil moisture content
3	DHT22 Sensor	Temperature & Humidity Sensor	Measures ambient temperature and humidity
4	Relay Module	1-Channel 5V Relay	Controls the water pump automatically
5	Mini Water Pump	DC Water Pump	Provides irrigation when soil moisture is low
6	OLED Display	0.96-inch I2C OLED Display	Displays real-time sensor data locally
7	Jumper Wires	Male-to-Male / Male-to-Female	Electrical interconnections between components
8	Breadboard	Standard Breadboard	Prototype circuit assembly
9	Laptop / Computer	USB Powered	Provides power, programming, and monitoring support

Table 3.1 – Hardware Components Used

3.4 : Software Components Used

Sl. No.	Software / Platform	Purpose
1	Arduino IDE	Writing, compiling, and uploading code to ESP32
2	Firebase	Cloud database for real-time data storage and monitoring
3	HiveMQ	MQTT broker for IoT communication and data transfer
4	FastAPI	Backend API development and data processing
5	React.js Dashboard	Real-time visualization and monitoring of sensor data
6	Google Gemini API (AI Module)	Generates intelligent irrigation and farming recommendations
7	Web Browser	Accessing and monitoring the cloud dashboard

Table 3.2 – Software Components Used

CHAPTER 4 : IMPLEMENTATION & RESULTS

4.1 Implementation Steps

The implementation of the Neuro VP Dhara Core system was carried out through a systematic integration of hardware and software components. The following steps were followed during the development process:

Step 1: Component Selection

The required hardware components such as ESP32, Capacitive Soil Moisture Sensor, DHT22 Sensor, 16×2 LCD Display with I2C Module, Relay Module, Water Pump, Breadboard, and Jumper Wires were selected based on project requirements.

Step 2: Circuit Assembly

All components were interconnected using a breadboard and jumper wires. The sensors, display module, relay, and water pump were connected to the ESP32 according to the designed circuit configuration.

Step 3: ESP32 Programming

The ESP32 was programmed using Arduino IDE. The program was developed to:

- Read soil moisture values.
- Measure temperature and humidity using DHT22.
- Display sensor readings on the LCD.
- Control the relay and water pump automatically.
- Transmit sensor data through Wi-Fi for cloud monitoring.

Step 4: Sensor Calibration

The soil moisture sensor was calibrated by observing readings under dry and moist soil conditions. Threshold values were determined to ensure accurate pump activation and deactivation.

Step 5: LCD Display Integration

The 16×2 LCD display was interfaced with the ESP32 using the I2C communication protocol. Real-time temperature, humidity, and pump status information were displayed continuously.

Step 6: Relay and Pump Testing

The relay module was connected to the water pump and tested under different soil moisture conditions. The pump was configured to switch ON when moisture levels dropped below the predefined threshold and switch OFF when sufficient moisture was detected.

Step 7: Cloud Connectivity

The ESP32 was connected to a Wi-Fi network to enable communication with cloud services such as Firebase and HiveMQ. Sensor data was uploaded periodically for remote monitoring.

Step 8: Dashboard Development

A web-based dashboard was developed using React.js to visualize real-time sensor readings, irrigation status, and historical data.

Step 9: AI Recommendation Integration

Collected environmental data was processed through the AI recommendation module to generate intelligent irrigation suggestions and improve water management decisions.

Step 10: System Testing and Validation

The complete system was tested under different environmental conditions. Sensor readings, pump response, cloud communication, and dashboard performance were monitored to verify proper operation and reliability.

Step 11: Performance Evaluation

The system performance was evaluated based on sensor accuracy, response time, automation efficiency, and cloud data transmission reliability. The obtained results demonstrated effective monitoring and automated irrigation control.

4.2 Photographs / Screenshots / Circuit / Model images

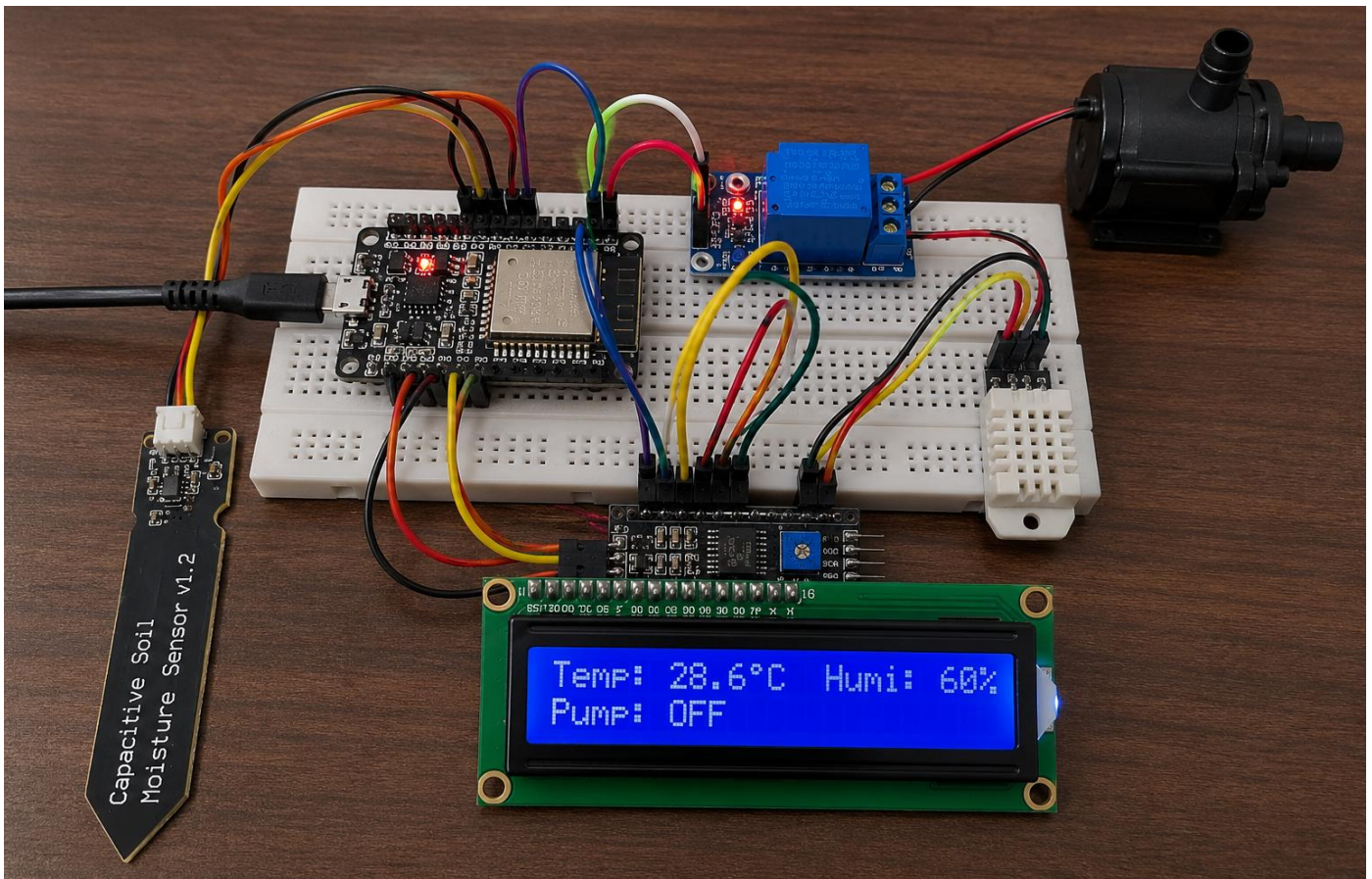


Figure 4.1 - Circuit Design

4.3 Observation

Parameter	Value
Location	Hoskote, Bengaluru Rural District, Karnataka, India
Latitude	13.0862° N
Longitude	77.7066° E
Area Surveyed	2.5 Acres (10,117 m ²)
Soil Type	Red Loamy Soil
Crop Type	Paddy Cultivation

Table 4.1 - Farmland Survey Information

Time	Soil Moisture (%)	Temperature (°C)	Humidity (%)	Pump Status
06:00 AM	68.5	24.6	78.2	OFF
08:00 AM	61.2	27.1	72.8	OFF
10:00 AM	52.7	30.4	66.5	OFF
12:00 PM	41.8	33.8	58.4	ON
02:00 PM	37.2	35.1	54.7	ON
04:00 PM	49.6	32.2	60.9	OFF

Table 4.2 - Sensor Readings and Irrigation Status

Condition	Moisture Range (%)	Pump Action
Moisture Sufficient	> 60	OFF
Moisture Normal	50 – 60	OFF
Moisture Low	< 45	ON
Moisture Restored	> 45 after irrigation	OFF

Table 4.3 - Irrigation Decision Summary

4.4 Results & Graphs

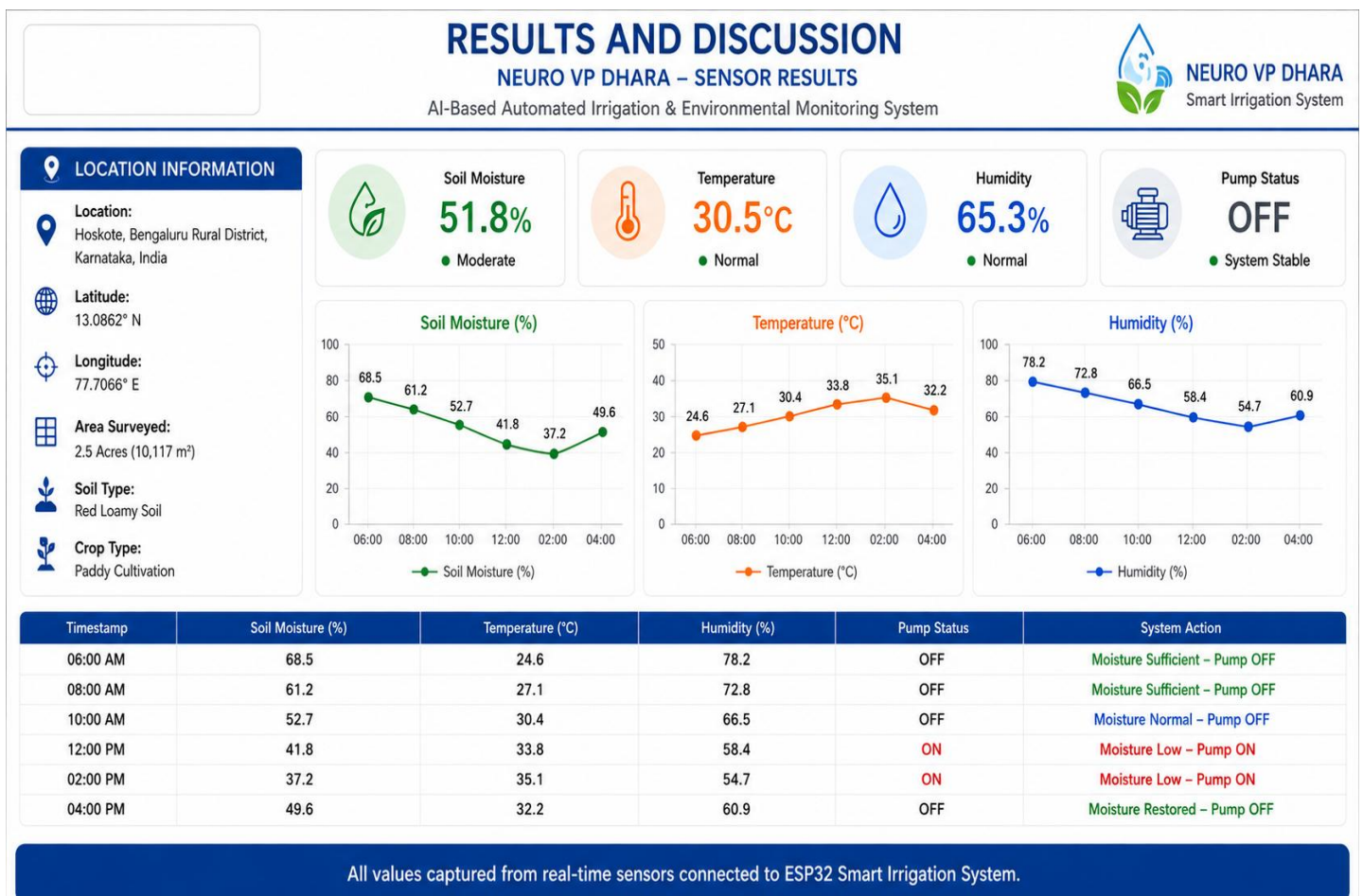


Figure 4.2 - Results and Discussion

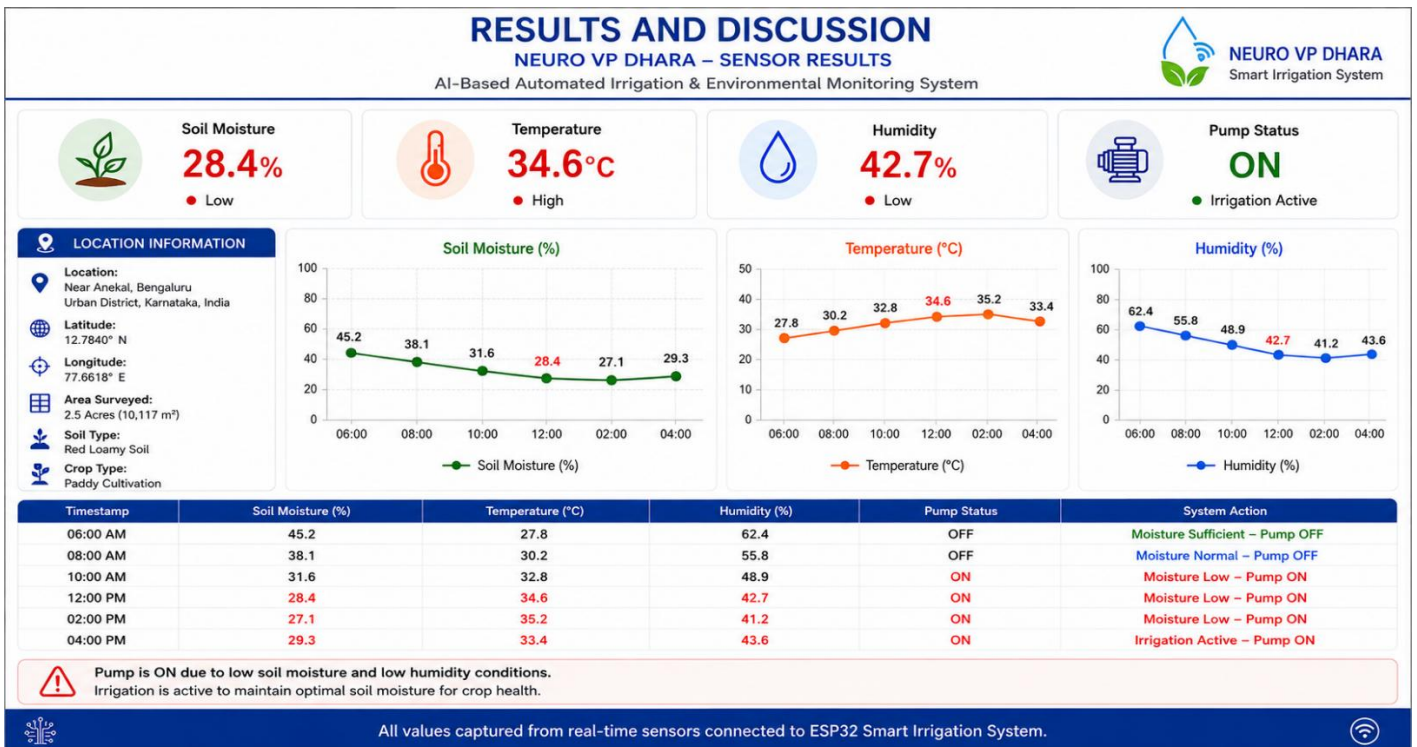


Figure 4.3 - Results and Discussion

4.5 Output & Web Interface



Figure 4.4 - Output & Web Interface(1)



Figure 4.5 - Output & Web Interface(2)

4.6 Limitation

- The system monitors only soil moisture, temperature, and humidity, while other important agricultural parameters are not considered.
- Real-time monitoring and cloud connectivity depend on the availability of a stable Wi-Fi connection.
- Sensor readings are collected from a single location and may not represent the entire field accurately.
- The prototype is powered through a laptop USB connection and is not intended for direct standalone field deployment.
- The irrigation system operates based on predefined moisture threshold values, which may need adjustment for different crops and soil conditions.

CHAPTER 5 : CONCLUSION AND FUTURE SCOPE

5.1 Summary of Work / Learning Outcomes

- Studied the importance of smart irrigation and IoT-based solutions in modern agriculture.
- Designed and developed an ESP32-based intelligent soil monitoring system for agricultural applications.
- Interfaced a capacitive soil moisture sensor and DHT22 sensor with the ESP32 microcontroller.
- Implemented automatic irrigation control using a relay module and water pump based on soil moisture levels.
- Established wireless communication for real-time monitoring and data transmission.
- Developed a dashboard for visualizing environmental and soil parameters.
- Learned the integration of hardware components, sensors, and cloud-based platforms in an IoT ecosystem.
- Gained practical knowledge of embedded programming using Arduino IDE and ESP32.
- Understood the application of automation techniques for efficient water management.
- Improved skills in circuit design, hardware interfacing, troubleshooting, and system testing.
- Analyzed sensor data and evaluated system performance under different environmental conditions.
- Demonstrated the potential of smart technologies in reducing water wastage and supporting sustainable agriculture.

5.2 Conclusion

The Neuro VP Dhara Core project successfully demonstrates the application of Internet of Things (IoT) technology in smart agriculture through real-time soil and environmental monitoring. The system utilizes an ESP32 microcontroller, capacitive soil moisture sensor, and DHT22 sensor to continuously monitor field conditions and automate irrigation based on soil moisture levels. By integrating sensing, automation, and cloud connectivity, the proposed solution helps reduce water wastage, minimize manual intervention, and improve irrigation efficiency.

The developed prototype effectively collects and processes sensor data, displays real-time information, and controls the water pump automatically whenever irrigation is required. The results obtained during testing confirm the reliability of the system in maintaining optimal soil moisture conditions. Overall, the project demonstrates a cost-effective and practical approach toward sustainable agriculture and highlights the potential of smart technologies in improving resource management and agricultural productivity.

5.3 Future Enhancement

- Integration of additional sensors such as pH, nutrient, and rainfall sensors for comprehensive soil analysis.
- Development of a dedicated mobile application for real-time monitoring and control.
- Incorporation of weather forecasting data to improve irrigation decision-making.
- Implementation of advanced AI and machine learning algorithms for predictive irrigation and crop recommendations.
- Deployment of solar-powered operation for standalone field usage.
- Expansion of the system to support multiple sensor nodes for large-scale farm monitoring.
- Integration of GPS and geolocation services for location-based agricultural analytics.
- Addition of alert and notification systems through SMS, email, or mobile applications.
- Enhancement of cloud storage and data analytics features for long-term agricultural insights.
- Development of a fully autonomous smart farming platform with minimal human intervention.

CHAPTER 6 : SDGs MAPPED

6.1 SDGs Mapped to the Project

6.1.1 SDG 7: Affordable and Clean Energy

The proposed system is designed using low-power IoT components such as the ESP32 microcontroller, making it energy-efficient and suitable for sustainable agricultural applications. By automating irrigation based on real-time soil moisture conditions, the system minimizes unnecessary operation of the water pump, thereby reducing energy consumption. The project demonstrates how smart technologies can optimize resource usage and contribute to energy-efficient farming practices.

6.1.2 SDG 9: Industry, Innovation and Infrastructure

The project integrates modern technologies such as the Internet of Things (IoT), cloud computing, wireless communication, and AI-based recommendations to develop an intelligent irrigation system. It promotes innovation in the agricultural sector by replacing conventional irrigation methods with a data-driven and automated approach. The system serves as an example of how advanced technological infrastructure can be utilized to improve agricultural productivity and support the development of smart farming solutions.

6.1.3 SDG 13: Climate Action

Efficient water management plays a significant role in addressing environmental challenges associated with agriculture. The proposed system continuously monitors soil and environmental conditions, ensuring that irrigation is performed only when necessary. This helps conserve water resources and supports sustainable agricultural practices. By reducing water wastage and encouraging responsible resource utilization, the project contributes to climate-resilient farming and aligns with global efforts toward environmental sustainability and climate action.

SDG No.	Sustainable Development Goal	Contribution of the Project
SDG 7	Affordable and Clean Energy	Reduces energy consumption through efficient and automated irrigation control using low-power IoT devices.
SDG 9	Industry, Innovation and Infrastructure	Utilizes IoT, cloud technologies, and AI to develop an innovative smart agriculture solution.
SDG 13	Climate Action	Promotes sustainable water management and supports environmentally responsible agricultural practices.

Table 6.1 – SDGs Mapped

CHAPTER 7 : REFERENCES

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CHAPTER 8 : APPENDIX

8.1 Key Components Used

Component	Key Specifications
ESP32 Development Board	ESP32-WROOM-32, Dual-Core Xtensa LX6 Processor, Up to 240 MHz Clock Frequency, 4 MB Flash Memory, 3.3 V Operating Voltage, Wi-Fi and Bluetooth Enabled, Supports UART, SPI, I2C, PWM, and ADC Communication Protocols
Capacitive Soil Moisture Sensor	Operating Voltage: 3.3 V–5 V, Analog Output, Corrosion Resistant Design, Used for Soil Moisture Measurement
DHT22 Sensor	Operating Voltage: 3.3 V–6 V, Temperature Range: -40°C to 80°C, Humidity 100% RH, Temperature Accuracy: $\pm 0.5^{\circ}\text{C}$, Humidity Accuracy: $\pm 2\%$ RH, Digital Output
16×2 LCD Display with I2C Module	16 Characters $\times$ 2 Lines Display, Operating Voltage: 5 V, I2C Communication Interface, LED Backlight, Used for Real-Time Data Display
Relay Module	1-Channel Relay, Operating Voltage: 5 V, Digital Trigger Input, Switching Capacity up to 10 A, Used for Automatic Pump Control
Mini Water Pump	DC Submersible Water Pump, Operating Voltage: 3 V–6 V, Current Consumption: 130–220 mA, Flow Rate: Approximately 80–120 L/hr, Controlled through Relay Module
Laptop USB Power Supply	Supply Voltage: 5 V DC, USB Powered, Provides Power to ESP32 and Connected Components during Development and Testing

Table 8.1 – Key Components Used

8.2 Abbreviations

Abbreviation	Full Form
IoT	Internet of Things
ESP32	Espress if 32-bit Microcontroller
LCD	Liquid Crystal Display
MQTT	Message Queuing Telemetry Transport
API	Application Programming Interface
AI	Artificial Intelligence
USB	Universal Serial Bus

Table 8.2 - Abbreviations

8.3 Project Outcomes

- Real-time monitoring of soil moisture, temperature, and humidity.
- Automated irrigation control based on soil moisture levels.
- Cloud-based monitoring and data visualization.
- AI-assisted recommendations for efficient irrigation management.
- Reduction in water wastage through intelligent decision-making.

8.4 Safety Precautions

- Ensure proper wiring before powering the circuit.
- Avoid exposing electronic components to water during testing.
- Verify voltage ratings before connecting sensors and modules.
- Disconnect power before modifying circuit connections.
- Handle the water pump carefully to prevent short circuits.